

Atropine sulfate

(Atropine)

I. Classification

- Parasympathetic blocking agent

II. Action(s)

- Inhibits parasympathetic actions of acetylcholine at postganglionic cholinergic neuroreceptor sites
- Decreases vagal tone, allows bronchial dilation, decreases secretions in respiratory tract and GI tract

III. Indication(s)

- Symptomatic bradycardia
- Organophosphate and nerve agent poisoning
- Management of Excessive Secretions (MIH Hospice / Palliative Care)

IV. Contraindications

- Known hypersensitivity
- Tachycardia
- No absolute contraindications in use for organophosphate / nerve agent poisoning

V. Precautions

- Can cause paradoxical bradycardia if pushed slowly or if less than therapeutic dose given (less than 0.5 mg in adults, less than 0.1 mg in children)

VI. Drug interactions

- Other anticholinergics – additive anticholinergic effects, vagal blockade
- Digitalis, cholinergic, neostigmine – potential adverse reaction
- Antihistamines, procainamide, quinidine, antipsychotics, antidepressants, benzos enhance effects

VII. Adult Dosing and Administration

- Symptomatic Bradycardia: 0.5 mg IV / IO (max dose 3 mg) – repeat per guideline
- Nerve agent / Organophosphate: Consult On-Call Medical Director if given
 - 2 – 4 mg IV / IO / IM, repeat as necessary until patient is asymptomatic, no max dose

VIII. Pediatric Dosing and Administration

- Symptomatic Bradycardia: 0.02 mg/kg IV/IO (Minimum dose 0.1 mg, Max single dose 0.5 mg, Max total dose 3 mg) – repeat per guideline
- Nerve agent / organophosphate: Consult On-Call Medical Director if given
 - 0.05 mg/kg IV/IM, repeat as necessary (max single dose 2mg, no max total dose for poisoning)

IX. Onset

- 2-5 minutes

X. Duration

- Half-life is approximately 3 hour

XI. Possible Adverse Effects

- Tachycardia, cardiac dysrhythmia
- Seizures
- Dry mouth, difficulty swallowing
- Flushed skin
- Dilated pupils / blurred vision
- Acute glaucoma

XII. Other Notes

- Not recommended in ASYMPTOMATIC bradycardia